$x[n]$ 周期 N

函数基底 $\{\phi[n] = e^{j2\pi mn/N}\}_{m=1,\cdots,N}$

1) 正交未归一化

$$\|\phi_m\|^2 = \langle \phi_m, \phi_m \rangle = \sum_{n=0}^{N-1} e^{j2\pi 0} = N$$

$$\langle \phi_m, \phi_k \rangle = \sum_{n=0}^{N-1} e^{-j2\pi mn/N} \cdot e^{j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} e^{j2\pi(k-m)n/N} = \frac{1-e^{j2\pi(k-m)}}{1-e^{j2\pi(k-m)/N}} = 0$$

2) 有限维

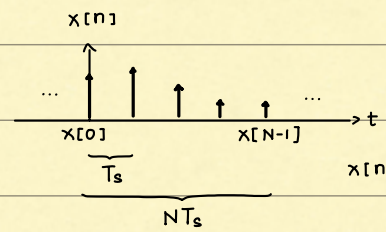
复指数函数仅 N 个，$\phi_m = \phi_{m+N}$

$$e^{j2\pi(m+N)n/N} = e^{j2\pi mn/N} \cdot e^{j2\pi n} = e^{j2\pi mn/N}$$

基频 $\omega_0 = \frac{2\pi}{N}$，周期为 N，$\frac{N}{2}$，$\cdots$，$\frac{N}{N}$ 再划细无意义

IDFT：DFS 级数 $x[n] = \sum_{m=0}^{N-1} y[m] e^{j2\pi mn/N}$

DFT：求系数 $y[m] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi mn/N}$



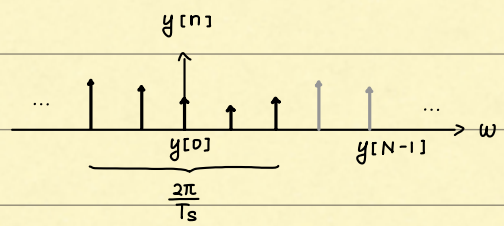
$x[n] = x(nT_s)$

$$x[n] = \sum_{m=0}^{N-1} y[m] \phi_m[n]$$

$$\langle \phi_m, x \rangle = \langle \phi_m, \phi_m \rangle y[m] = N y[m]$$

$$y[m] = \frac{1}{N} \langle \phi_m, x \rangle$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi mn/N}$$



与 $y(\omega)$ 比 $y[m] = y(\frac{2\pi}{NT_s} m)$

与 $y(f)$ 比 $y[m] = y(\frac{m}{NT_s})$

截止频率 $f_c = \frac{1}{2T_s}$，$m_c = \frac{1}{2}N$